QuantileMix: A Python package for quantile g-computation

Nilson Chapagain

Texas A&M University, United States

nchapagain@tamu.edu

Summary

Exposure-mixture studies ask how an outcome changes when several exposures increase together. Quantile g-computation defines this increase using an ordered score for each exposure, so measurements in different units can enter the same joint comparison (Keil et al., 2020). With four quantile groups, for example, scores range from 0 to 3, and the comparison increases every exposure score by one. The corresponding changes in the original measurement units can differ across exposures.

We developed QuantileMix to estimate these joint contrasts in Python, with confidence intervals and predictions based on the fitted scoring rule. The package combines exposure scoring with selected models for continuous, binary, count and survival outcomes. More restricted methods cover censored, multicategory and excess-zero outcomes. Fitting and prediction use NumPy and SciPy (Harris et al., 2020; Virtanen et al., 2020) without an R runtime. Worked examples, numerical comparisons with R and repeated-dataset Gaussian studies document what has been checked and where uncertainty remains.

Motivation

A mixture analysis requires both a joint effect estimate and an estimate of its uncertainty. In an additive model, the joint effect is a sum of exposure coefficients, but its variance also depends on covariances between their estimates (Keil et al., 2020). Researchers also need predictions that use the same exposure scale as the fitted model. Recalculating quantile boundaries on new data would change that scale.

QuantileMix brings these calculations into a Python workflow. It stores fitted cutpoints, calculates method-specific joint contrasts and uncertainty, and retains the information needed for prediction. This lets researchers carry the same fitted analysis from exposure scoring to results, while checking the assumptions and limits of the chosen method.

Related software

The R package qgcomp provides an established and broader set of quantile g-computation methods (Keil, 2026). We use selected calculations from qgcomp as numerical references. QuantileMix is a separate, unofficial Python implementation with fewer functions and options; it does not introduce the underlying statistical method.

Within Python, Statsmodels provides regression models and linear contrasts (Seabold and Perktold, 2010). DeepMix combines additive quantile g-computation with fitted cutpoints and prediction (Kang, 2026). At the cited source revision, its Python result does not include a coefficient covariance matrix, standard error or confidence interval. QuantileMix combines these uncertainty calculations with joint contrasts and prediction, and documents the effect scale returned by each supported method. Its contribution is the software and reproducible evidence for the selected calculations.

Methods and implementation

We separate exposure scoring, model fitting and results so that a fitted analysis can be reused without changing its scoring rule. Results retain estimates, covariance information, diagnostics and prediction inputs.

Exposure scoring

For person i and exposure j, let X_{ji} denote the measured exposure and X_{ji}^q its quantile score, following Keil et al. (2020). When quantization is requested, fitting determines the cutpoints from the training data unless cutpoints are supplied. Prequantized inputs use the supplied scores directly. With four groups, $X_{ji}^q \in \{0, 1, 2, 3\}$. Tied measurements can produce groups of unequal size. We save the fitted cutpoints and apply them to new observations; we do not estimate a new set of cutpoints for prediction.

Figure 1 shows this step for one exposure in a saved Gaussian example. The training observations determine the boundaries in panel (a), and panel (b) applies those same boundaries to new values. A one-score increase therefore has the same fitted definition in training and prediction, although it need not represent a fixed change in the original measurement units.

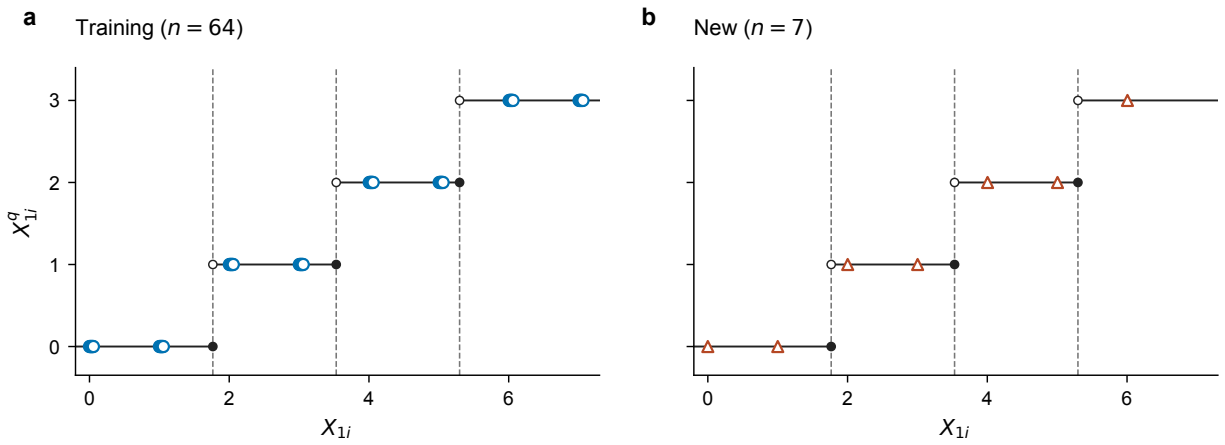


Figure 1: Exposure scores from a fixed set of training cutpoints. (a) The first exposure among 64 training observations. (b) Seven new exposure values scored with the same cutpoints. Vertical lines mark 1.76575, 3.5315 and 5.29725 in the example's exposure units; points show the saved score assignments, with nearby training points overlapping. Filled endpoints include a cutpoint; open endpoints exclude it. The step function is the fitted scoring rule, not an outcome-response curve.

Joint effect estimation

Once exposures are scored, the additive Gaussian model gives a simple joint contrast. Let Y_i be the outcome, d the number of exposures, β_0 the intercept, β_j an exposure coefficient and ϵ_i the random error. Following Keil et al. (2020), omitting adjustment variables for clarity and taking $E(\epsilon_i \mid X_{1i}^q, \ldots, X_{di}^q) = 0$, we write

$$Y_i = \beta_0 + \sum_{j=1}^{d} \beta_j X_{ji}^q + \epsilon_i, \qquad \psi = \sum_{j=1}^{d} \beta_j.$$

Increasing every exposure score by one changes the model mean by the sum ψ. Adjustment variables, when included, are held fixed. Formula routines can represent categorical adjustment variables through indicator terms. The comparison requires support for both exposure combinations in the study population (Keil et al., 2020). We estimate the contrast by $\widehat{\psi} = \sum_{j=1}^{d} \widehat{\beta}_j$ and retain all covariance terms when estimating its variance:

$$\widehat{\mathrm{Var}}(\widehat{\psi}) = \sum_{j=1}^{d} \sum_{k=1}^{d} \widehat{\mathrm{Cov}}(\widehat{\beta}_j, \widehat{\beta}_k).$$

The covariance terms account for how coefficient estimates vary together; adding only their variances would generally give a different standard error (Keil et al., 2020). Figure 2 shows why individual coefficients and their joint contrast also need separate interpretation: coefficients with opposite signs can offset one another. For a nonzero fitted coefficient $\widehat{\beta}_k$, its sign weight is $w_k = |\widehat{\beta}_k| / \sum_{j:\, \widehat{\beta}_j \widehat{\beta}_k > 0} |\widehat{\beta}_j|$. The denominator sums absolute coefficients with the same sign, so positive and negative contributions are summarized separately (Keil et al., 2020; Keil, 2026). These weights describe the fitted model rather than causal importance.

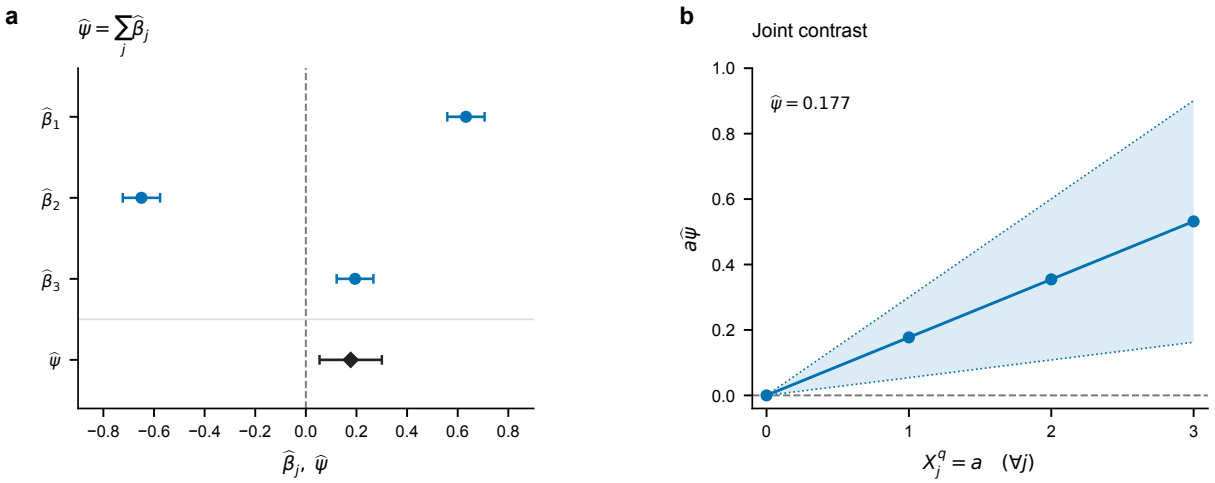


Figure 2: Estimated coefficients and joint exposure contrast in an additive Gaussian model. (a) $\widehat{\beta}_j$ and $\widehat{\psi} = \sum_{j=1}^{3} \widehat{\beta}_j$. (b) Estimated mean difference $a\widehat{\psi}$ between $X_j^q = a$ and $X_j^q = 0$ for all j, with $a \in \{0, 1, 2, 3\}$. Bars in (a) and shading in (b) denote pointwise 95% normal confidence intervals. Shading is not a prediction interval for an individual outcome. The synthetic example has $n = 64$, $d = 3$, and two adjustment variables.

Supported models and prediction

The coefficient sum above applies to the additive Gaussian mean. For other outcomes, the effect scale and the way predictions are averaged must also be specified. QuantileMix therefore provides separate functions for conditional and marginal binary and count effects. Conditional effects hold adjustment variables fixed; marginal effects average over their distribution. A conditional odds ratio can differ from a marginal odds ratio even without confounding (Greenland et al., 1999).

For marginal estimation, we predict outcomes at specified exposure settings, average over the observed adjustment values, and summarize the predictions with the specified marginal model (Keil et al., 2020; Keil, 2026). Nonlinear terms and interactions require method-specific contrasts rather than an automatic sum of coefficients. Bootstrap and estimating-equation methods have separate limits on nonlinear terms and clusters (Keil, 2026). A row bootstrap resamples independent observations with replacement (Efron, 1979); it does not preserve within-cluster dependence.

The prediction scale depends on the outcome model. Figure 3 shows conditional means, probabilities, a latent mean before censoring and Cox survival curves (Cox, 1972) in separate synthetic examples. The panels illustrate the outputs; their different vertical scales are not a comparison of model performance.

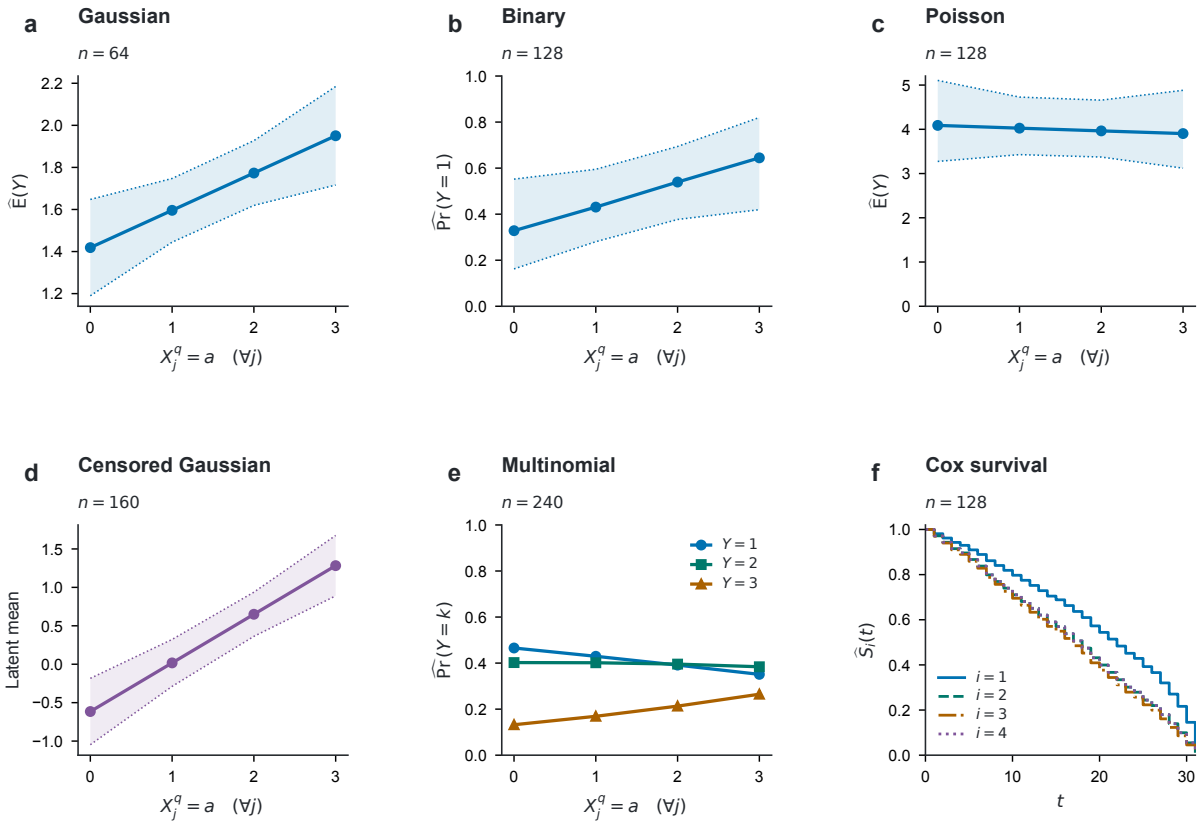


Figure 3: Conditional predictions for six outcome models. (a, c) $\widehat{E}(Y)$ under Gaussian and Poisson models; (b) $\widehat{\Pr}(Y = 1)$; (d) the latent Gaussian mean before censoring; (e) $\widehat{\Pr}(Y = k)$, $k = 1, 2, 3$ (low, mid, high); (f) Cox survival estimates $\widehat{S}_i(t)$ for four observed profiles i. In (a–e), $X_j^q = a$ for all j and adjustment variables are zero; lines connect the discrete score levels. Shading in (a–d) denotes pointwise 95% Wald confidence intervals, with binary and Poisson limits transformed from the linear predictor scale. No intervals are shown in (e–f). Each panel uses separate synthetic data; t is in the example's time units.

For excess-zero counts, the selected zero-inflated and hurdle models return probabilities for individual count values (Zeileis et al., 2008). Figure 4 compares these fitted probabilities at two common exposure scores within each example. Probability outside the displayed count range is retained, so the plotted values are not rescaled to sum to one.

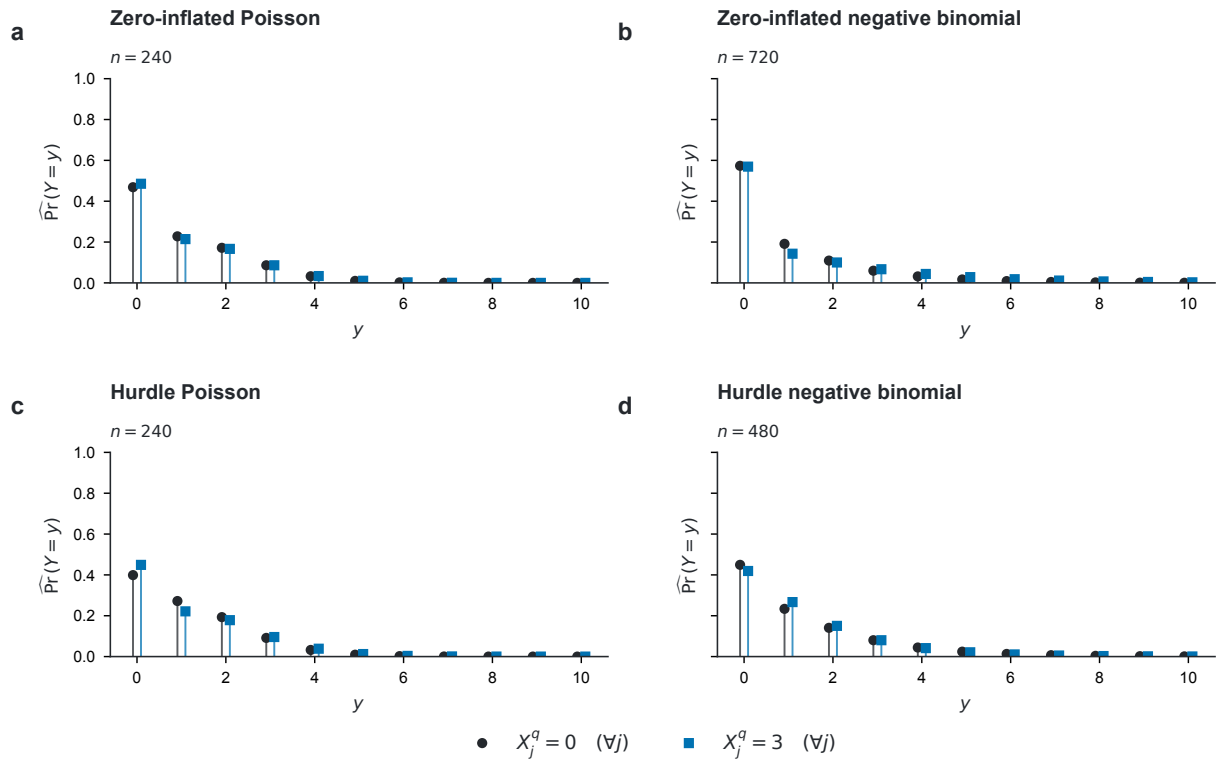


Figure 4: Fitted probability mass functions for two-part count models. (a) Zero-inflated Poisson; (b) zero-inflated negative binomial; (c) hurdle Poisson; (d) hurdle negative binomial. Points show $\widehat{\Pr}(Y = y)$, $y = 0, \ldots, 10$, at $X_j^q = 0$ or $X_j^q = 3$ for all j, with adjustment variables zero. Probabilities are not renormalized over the displayed counts; $\widehat{\Pr}(Y > 10)$ is retained in the source table. Each panel uses separate synthetic data. No confidence intervals are shown.

Matplotlib supplies the plots (Hunter, 2007). The manual gives supported formula expressions and distinguishes case weights used in fitting from sign weights used to describe exposure coefficients. It also states which methods accept cluster identifiers.

Interpretation and assumptions

A causal interpretation requires well-defined exposure interventions. The outcome under a person's observed exposure must equal the observed outcome (consistency), and one person's exposure must not affect another person's outcome (no interference). After adjustment, no unmeasured common causes of exposure and outcome may remain. The compared exposure combinations must have support at the adjustment values used (positivity), and the outcome model must be correctly specified (Keil et al., 2020). The software cannot verify these assumptions.

Missing-data handling also affects which observations enter the analysis. Missing values are rejected by default; explicit row removal is available where documented. Removing incomplete rows does not by itself prevent bias (Sterne et al., 2009).

Validation and limitations

Numerical agreement and interval coverage

We use two kinds of checks because numerical agreement and repeated-dataset performance answer different questions. A comparison with R tests whether selected software calculations agree. A repeated-dataset study examines how an estimator and its intervals behave under the specified data-generating model. Our validation records provide the settings, scripts and results for both.

We compared nine Gaussian cases with installed R qgcomp 2.19.6. In a separate 24,000-dataset Gaussian study, we compared Python, installed R and an independent calculation using QR factorization. All fits succeeded, and all paired comparisons met the recorded numerical tolerances. The 12 conditions used $d = 4$

exposures, $n \in \{100, 500\}$ and 2,000 datasets per condition ($12 \times 2{,}000 = 24{,}000$). Across these conditions, 94.55%–96.10% of the nominal 95% Python/R intervals contained the true effect. These intervals use normal critical values rather than exact Gaussian Student-t limits.

Bias, standard errors and bootstrap intervals

We evaluated Gaussian row-bootstrap intervals in another 24,000-dataset study, using the same 12-condition design. Within that study, model-based and bootstrap methods use the same datasets and the same point estimate; their standard errors and intervals differ. Figure 5 separates the bias of $\widehat{\psi}$ from the calibration of its reported standard errors. Bias is the mean of $\widehat{\psi} - \psi$ across datasets. Its Monte Carlo standard error measures uncertainty from using a finite number of datasets, rather than uncertainty about ψ in one fitted dataset.

The second column compares the root mean square of the reported standard errors with the empirical standard deviation of $\widehat{\psi}$. A ratio near one indicates agreement between these two summaries. Across the 12 conditions, all pointwise Monte Carlo intervals for bias include zero, and the standard-error ratios range from 0.972 to 1.038. These summaries do not ensure nominal coverage in every condition, which we assess separately in Figure 6.

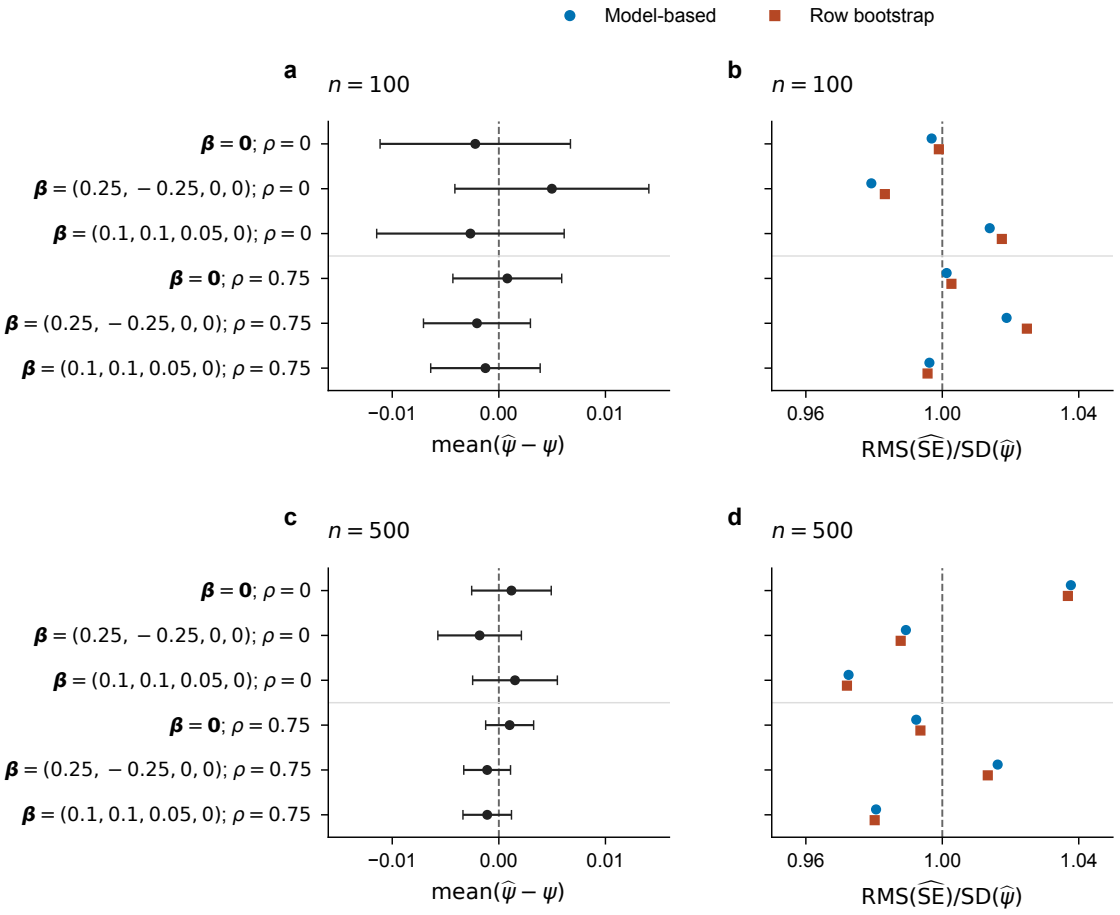


Figure 5: Bias and standard-error calibration in the saved Gaussian bootstrap study. (a, b) $n = 100$; (c, d) $n = 500$. Each condition contains 2,000 datasets. Left: mean $\widehat{\psi} - \psi$ with normal 95% Monte Carlo intervals (± 1.96 Monte Carlo standard errors). The shared point estimate is plotted once. Right: root mean square of reported standard errors divided by the empirical standard deviation of $\widehat{\psi}$, for model-based and row-bootstrap methods; no uncertainty intervals are shown for these ratios. Dashed lines mark zero bias and a ratio of one. Rows identify $\boldsymbol{\beta}$ and the residual latent exposure correlation ρ. All fits succeeded; Monte Carlo intervals are pointwise, not simultaneous across conditions.

The bootstrap normal Wald intervals are approximately $\widehat{\psi} \pm 1.96\sqrt{\widehat{\mathrm{Var}}_{\mathrm{boot}}(\widehat{\psi})}$, where $\widehat{\mathrm{Var}}_{\mathrm{boot}}(\widehat{\psi})$ is the sample variance of 200 bootstrap estimates. These are not percentile intervals. All fits succeeded. In three conditions, observed coverage was below 95%, with pointwise 95% Wilson intervals lying entirely below 95%. These comparisons were not adjusted for testing multiple conditions.

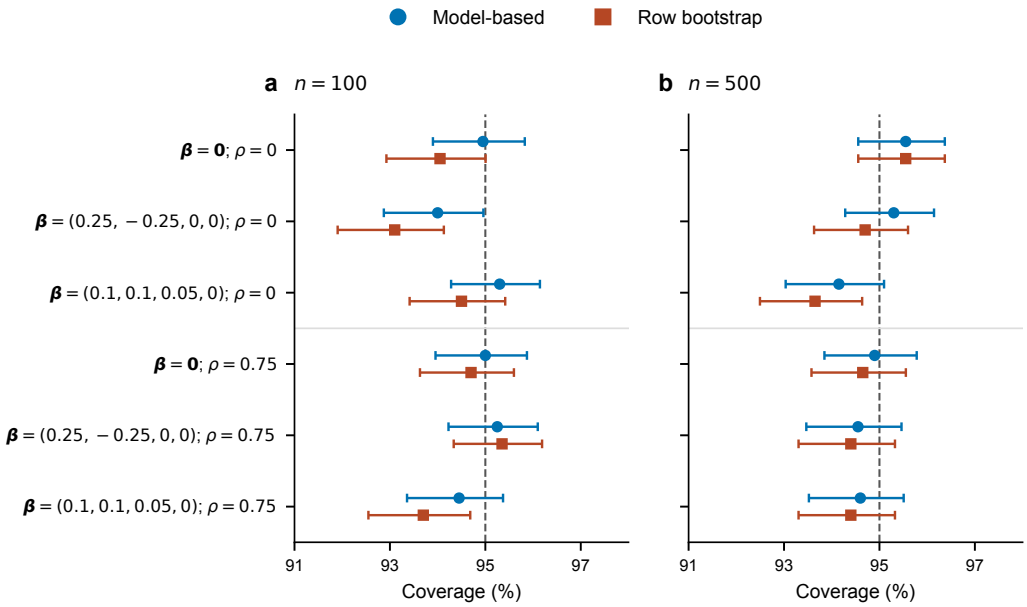


Figure 6: Empirical coverage of nominal 95% Gaussian confidence intervals. (a) $n = 100$; (b) $n = 500$. Points show $\frac{100}{2{,}000} \sum_{r=1}^{2{,}000} \mathbf{1}\{\psi \in I_r\}$, where I_r is the confidence interval from dataset r and $\mathbf{1}$ is the indicator function. Bars denote pointwise 95% Wilson Monte Carlo intervals (Wilson, 1927); the dashed line marks 95% coverage. Rows identify $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_4)$ and residual latent exposure correlation ρ. Model-based and row-bootstrap methods use the same 2,000 datasets per condition (24,000 total), with no fit failures. Bootstrap intervals use normal critical values and standard errors from 200 resamples with fixed exposure scores. The Wilson intervals are neither simultaneous intervals across conditions nor intervals for paired coverage differences.

Both studies used known population scores and correctly specified Gaussian models. Their coverage results therefore do not establish performance when cutpoints are estimated, the outcome model is incorrect, or other families are fitted. Numerical agreement and these Gaussian studies support the specific calculations and conditions tested; neither establishes causality.

Reproducible materials and limits

The repository includes study settings, scripts, results, reference comparisons and worked examples so that others can inspect and repeat the calculations. We have not yet documented use by other researchers or a completed real-data research application. These remain separate steps from software agreement and Gaussian calibration. The current evidence does not establish widespread use or complete R-package coverage.

AI usage disclosure

ChatGPT was used as an AI assistant. AI-assisted language editing was used to improve wording in parts of the manuscript. All code and text generated by AI was proofread by the authors. AI-generated code was also extensively tested.

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